

Instalacion y uso de Haskell

Nombre de la materia: Paradigmas de la Programacion

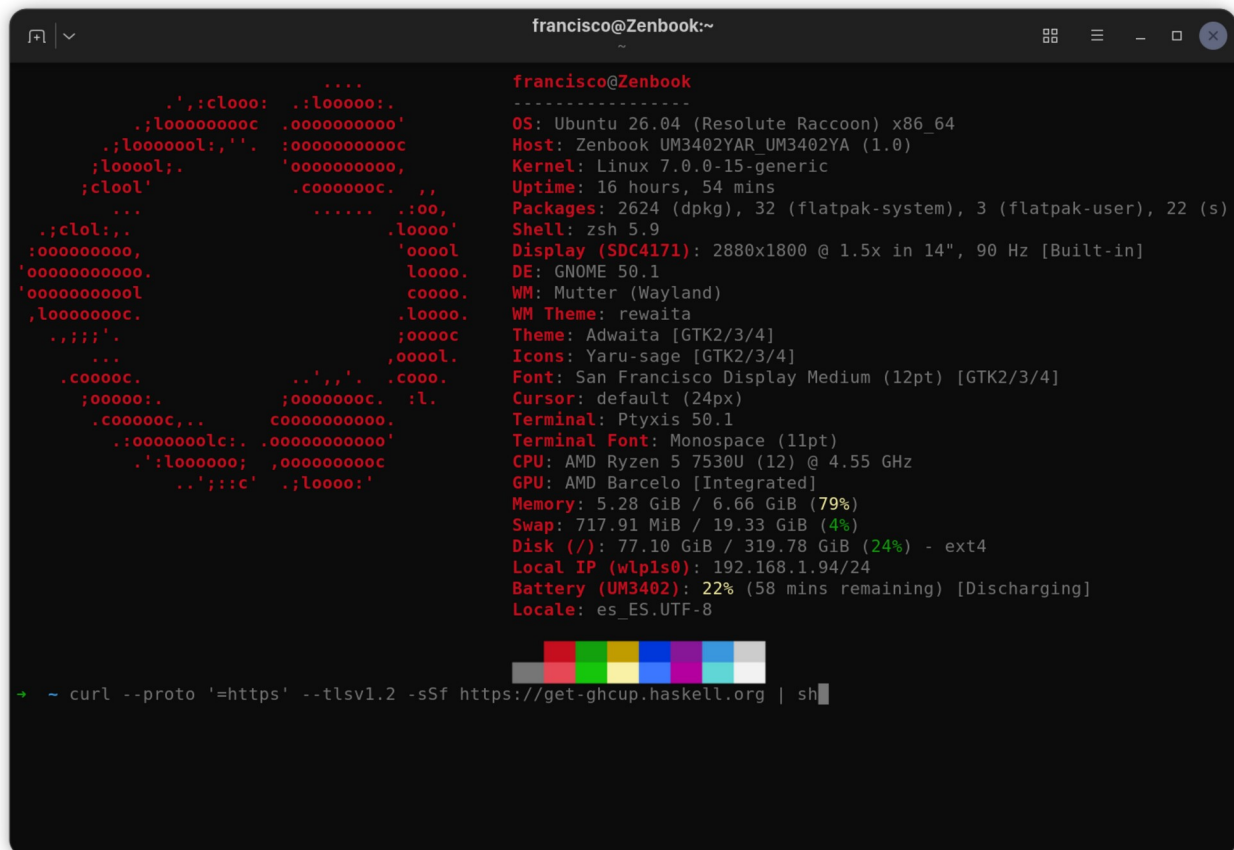
Institución: Universidad Autónoma de Baja California Sur (UABCS)

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Fecha de entrega: 10/05/2026

Paso 1 — Instalar GHCup (el gestor oficial de Haskell)



```
francisco@Zenbook:~  
francisco@Zenbook  
-----  
OS: Ubuntu 26.04 (Resolute Raccoon) x86_64  
Host: Zenbook UM3402YAR UM3402YA (1.0)  
Kernel: Linux 7.0.0-15-generic  
Uptime: 16 hours, 54 mins  
Packages: 2624 (dpkg), 32 (flatpak-system), 3 (flatpak-user), 22 (s)  
Shell: zsh 5.9  
Display (SDC4171): 2880x1800 @ 1.5x in 14", 90 Hz [Built-in]  
DE: GNOME 50.1  
WM: Mutter (Wayland)  
WM Theme: rewaita  
Theme: Adwaita [GTK2/3/4]  
Icons: Yaru-sage [GTK2/3/4]  
Font: San Francisco Display Medium (12pt) [GTK2/3/4]  
Cursor: default (24px)  
Terminal: Ptyxis 50.1  
Terminal Font: Monospace (11pt)  
CPU: AMD Ryzen 5 7530U (12) @ 4.55 GHz  
GPU: AMD Barcelo [Integrated]  
Memory: 5.28 GiB / 6.66 GiB (79%)  
Swap: 717.91 MiB / 19.33 GiB (4%)  
Disk (/): 77.10 GiB / 319.78 GiB (24%) - ext4  
Local IP (wlp1s0): 192.168.1.94/24  
Battery (UM3402): 22% (58 mins remaining) [Discharging]  
Locale: es_ES.UTF-8  
  
→ ~ curl --proto '=https' --tlsv1.2 -sSf https://get-ghcup.haskell.org | sh
```



Paso 2 — Agregar GHCup al PATH

```
francisco@Zenbook:~  
      .'.:clooo: .:looooo:.  
      .:loooooooooooc .oooooooooooo'  
      .:loooooool:.'. :ooooooooooooc  
      ;loool;. 'oooooooooooo,  
      ;clool' .cooooooc. ,,  
      ...          ..... :oo,  
      .;clol:,. .loooo'  
      :ooooooooo, 'ooooo'  
      'oooooooooooo. loooo.  
      'ooooooooool coooo.  
      ,loooooooc. .loooo.  
      .,:;;'. ;ooooo.  
      ...          ,ooooo.  
      .cooooooc. .',,'.' .cooo.  
      ;oooo:. ;oooooooooc. :l.  
      .cooooooc,.. cooooooooooooo.  
      .:ooooooooolc:. :oooooooooooo'  
      .':loooooo; ,ooooooooooooc  
      ..':;c' ;loooo:'
```

```
francisco@Zenbook  
-----  
OS: Ubuntu 26.04 (Resolute Raccoon) x86_64  
Host: Zenbook UM3402YAR_UM3402YA (1.0)  
Kernel: Linux 7.0.0-15-generic  
Uptime: 44 mins  
Packages: 2626 (dpkg), 32 (flatpak-system), 3 (flatpak-user), 22 (snap)  
Shell: zsh 5.9  
Display (SDC4171): 2880x1800 in 14", 90 Hz [Built-in]  
DE: GNOME 50.1  
WM: Mutter (Wayland)  
WM Theme: rewaita  
Theme: Adwaita [GTK2/3/4]  
Icons: Yaru-sage [GTK2/3/4]  
Font: San Francisco Display Medium (12pt) [GTK2/3/4]  
Cursor: default (32px)  
Terminal: Ptyxis 50.1  
Terminal Font: Monospace (11pt)  
CPU: AMD Ryzen 5 7530U (12) @ 4.55 GHz  
GPU: AMD Barcelo [Integrated]  
Memory: 4.45 GiB / 6.66 GiB (67%)  
Swap: 523.21 MiB / 19.33 GiB (3%)  
Disk (/): 87.55 GiB / 319.78 GiB (27%) - ext4  
Local IP (wlp1s0): 10.200.11.162/24  
Battery (UM3402): 70% (5 hours, 38 mins remaining) [Discharging]  
Locale: es_ES.UTF-8
```



Paso 3 — Instalar GHC (el compilador)

The image shows a terminal window titled "francisco@Zenbook:~/Escriptorio". The terminal displays two columns of ASCII art. The left column features a large, stylized letter 'C' composed of dots and dashes. The right column contains system information for Ubuntu 26.04 (Resolute Raccoon) x86_64, including host name, kernel version, uptime, packages, shell, display resolution, desktop environment, window manager, theme, icons, font, cursor, terminal type, terminal font, CPU, GPU, memory usage, swap usage, disk usage, local IP, battery status, and locale. At the bottom left, a green prompt character followed by the word "Escriptorio" indicates a command prompt where the user has entered "ghcup install ghc 9.6.7" and "ghcup set ghc 9.6.7". A color calibration bar is visible at the bottom center.



Paso 4 — Verificar la instalación

The image shows a terminal window with three tabs at the top: "francisco@Zenbook:~", "francisco@Zenbook:~/Escritorio", and "francisco@Zenbook:~/Escritorio". The first tab displays a large ASCII art drawing of a cat's face made of dots and semicolons. The second and third tabs show the output of the command "cat /etc/os-release", which lists system details such as OS version (Ubuntu 26.04), host name (Zenbook), kernel version (Linux 7.0.0-15-generic), uptime (44 mins), packages (2626 dpkg, 3 flatpak-system, 3 flatpak-user, 22 s), shell (zsh 5.9), display resolution (2880x1800), DE (GNOME 50.1), WM (Mutter), WM Theme (rewaita), Theme (Adwaita), Icons (Yaru-sage), Font (San Francisco Display Medium), Cursor (default), Terminal (Ptyxis), Terminal Font (Monospace), CPU (AMD Ryzen 5 7530U @ 4.55 GHz), GPU (AMD Barcelo), Memory (4.46 GiB / 6.66 GiB at 67%), Swap (515.38 MiB / 19.33 GiB at 3%), Disk (/) (87.54 GiB / 319.78 GiB at 27% - ext4), Local IP (wlp1s0: 10.200.11.162/24), Battery (UM3402: 70% remaining), and Locale (es_ES.UTF-8). At the bottom left, there is a prompt "→ Escritorio ghc --version" followed by a cursor. To the right of the prompt is a color calibration chart consisting of a row of nine colored squares: black, red, green, yellow, cyan, blue, magenta, white, and grey.

Paso 5 — Instalar dependencia faltante



DASC

```
francisco@Zenbook:~  
francisco@Zenbook:~  
francisco@Zenbook:~/Escritorio
```

```
      .',;cloo:  .:looooo:  ....  
      .;loooooooc .oooooooc'  .  
      .;looooool:,'. :oooooooc  
      ;loool;.      'oooooooc,  
      ;clood'      .cooooooc.  ,,  
      ...          ..... :oo,  
      .;clol:,,      .loooo'  
      :oooooooc,      'ooool  
      'oooooooc.      loooo.  
      'ooooooool      coooo.  
      ,looooooc.      .loooo.  
      .,;;;'      ;oooc  
      ...          ,ooool.  
      .cooooc.      ..',,'. .cooo.  
      ;oooo:..      ;oooooc. :l.  
      .cooooooc,,  cooooooooc.  
      .:oooooolc:.. :oooooooc'  
      .':looooo; ,oooooooc  
      ..':;c' .;loooo:'
```

```
francisco@Zenbook  
-----  
OS: Ubuntu 26.04 (Resolute Raccoon) x86_64  
Host: Zenbook UM3402YAR_UM3402YA (1.0)  
Kernel: Linux 7.0.0-15-generic  
Uptime: 44 mins  
Packages: 2626 (dpkg), 32 (flatpak-system), 3 (flatpak-user), 22 (s)  
Shell: zsh 5.9  
Display (SDC4171): 2880x1800 in 14", 90 Hz [Built-in]  
DE: GNOME 50.1  
WM: Mutter (Wayland)  
WM Theme: rewaita  
Theme: Adwaita [GTK2/3/4]  
Icons: Yaru-sage [GTK2/3/4]  
Font: San Francisco Display Medium (12pt) [GTK2/3/4]  
Cursor: default (32px)  
Terminal: Ptyxis 50.1  
Terminal Font: Monospace (11pt)  
CPU: AMD Ryzen 5 7530U (12) @ 4.55 GHz  
GPU: AMD Barcelo [Integrated]  
Memory: 4.46 GiB / 6.66 GiB (67%)  
Swap: 515.38 MiB / 19.33 GiB (3%)  
Disk (/): 87.54 GiB / 319.78 GiB (27%) - ext4  
Local IP (wlp1s0): 10.200.11.162/24  
Battery (UM3402): 70% (6 hours, 12 mins remaining) [Discharging]  
Locale: es_ES.UTF-8
```

```
→ Escritorio sudo apt install libgmp-dev
```



DASC

Paso 6 — Crear el archivo

```
francisco@Zenbook:~/Escritorio
francisco@Zenbook:~
francisco@Zenbook:~/Escritorio

.,;clooo: .:looooo:.
.;loooooooc .oooooocccc'
.;looooool:.' .oooooocccc
;looooo;. 'oooooocccc,
;clool' .cooooooc.
...
.;clol:.. .:oo,
:ooooooccc, .loooo'
'oooooocccc. loooo.
'oooooocccc. coooo.
,looooooc. .loooo.
.,;:.' ;ooooc
... .oooo.
.coooo. .:,'. .cooo.
;oooo:.. ;ooooooc. :l.
.coooooc,. coooooocccc.
.:ooooooclc:.. .oooooocccc'
.';looooo; .oooooocccc
..;:c' .;loooo:'

francisco@Zenbook
-----
OS: Ubuntu 26.04 (Resolute Raccoon) x86_64
Host: Zenbook UM3402YAR_UM3402YA (1.0)
Kernel: Linux 7.0.0-15-generic
Uptime: 44 mins
Packages: 2626 (dpkg), 32 (flatpak-system), 3 (flatpak-user), 22 (s)
Shell: zsh 5.9
Display (SDC4171): 2880x1800 in 14", 90 Hz [Built-in]
DE: GNOME 50.1
WM: Mutter (Wayland)
WM Theme: rewaita
Theme: Adwaita [GTK2/3/4]
Icons: Yaru-sage [GTK2/3/4]
Font: San Francisco Display Medium (12pt) [GTK2/3/4]
Cursor: default (32px)
Terminal: Ptyxis 50.1
Terminal Font: Monospace (11pt)
CPU: AMD Ryzen 5 7530U (12) @ 4.55 GHz
GPU: AMD Barcelo [Integrated]
Memory: 4.46 GiB / 6.66 GiB (67%)
Swap: 515.38 MiB / 19.33 GiB (3%)
Disk (/): 87.54 GiB / 319.78 GiB (27%) - ext4
Local IP (wlp1s0): 10.200.11.162/24
Battery (UM3402): 70% (6 hours, 12 mins remaining) [Discharging]
Locale: es_ES.UTF-8

→ Escritorio nano ~/Escritorio/suma.hs
```



DASC

Con el contenido:

```
francisco@Zenbook:~/Escritorio
francisco@Zenbook:~
francisco@Zenbook:~/Escritorio

.;loooooooooo. .oooooooooooo'
.;loooooool:,'. :ooooooooooooo
;loool;. 'oooooooooooo,
;clool' .cooooooooooc. ,,
... .. .:oo,
.;clol:,. .looooo'
:oooooooooooo. 'ooooo
'oooooooooooo. loooo.
'oooooooooooo. coooo.
,looooooooooc. .loooo.
.,;:'. ;ooooo
... ,ooooo.
.cooooo. ..',,. .cooo.
;oooo:. ;oooooooooc. :l.
.cooooo,. cooooooooooooo.
.:ooooooooolc:. .oooooooooooo'
.';looooo; ,oooooooooooo
..'::c' .;loooo:'

OS: Ubuntu 26.04 (Resolute Raccoon) x86_64
Host: Zenbook UM3402YAR UM3402YA (1.0)
Kernel: Linux 7.0.0-15-generic
Uptime: 44 mins
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Cursor: default (32px)
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CPU: AMD Ryzen 5 7530U (12) @ 4.55 GHz
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Memory: 4.46 GiB / 6.66 GiB (67%)
Swap: 515.38 MiB / 19.33 GiB (3%)
Disk (/): 87.54 GiB / 319.78 GiB (27%) - ext4
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Battery (UM3402): 70% (6 hours, 12 mins remaining) [Discharging]
Locale: es_ES.UTF-8

+ Escritorio sumar :: Int -> Int -> Int
sumar x y = x + y

main :: IO ()
main = do
  putStrLn "Probando la función sumar:"
  print (sumar 5 3)
  print (sumar 10 20)
  print (sumar 100 (-45))
```

Paso 7 — ejecución y resultado:

```
francisco@Zenbook:~/Escritorio
francisco@Zenbook:~
francisco@Zenbook:~/Escritorio

→ Escritorio cd ~/Escritorio
ghc suma.hs -o suma
./suma
Probando la función sumar:
8
30
55
→ Escritorio
```

Archivos creados:

